MGM Macau and Cotai

Resource Efficiency Management Report

February 2022

Energenz Consulting Limited

EXECUTIVE SUMMARY

Energenz has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. Energenz appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

Key Performance Indicators

Energenz reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as Energenz's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to February 2022	2023	2024	2025
Macau	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-5.9%**	**-4.8%**	-	-	**-**
Cotai	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-6.8%**	**-7.5%**	-	-	**-**

**MGM Cotai ISO energy performance: e-bus consumption is included.*

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2022.

Table 2. Internal Actual Energy Performance Targets for 2025

MGM Properties	Baseline	Actual Saving Target[1]	Savings Achieved Up to February 2022
Macau	2019	15%	**-4.1%**
Cotai	2019	15%	**-5.6%**

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams.

Highlighted Changes in Operation

Macau

1. The chilled water supply temperature setpoint has been finetuned from the temperature range of 8°C – 9°C to 8.5°C – 9.5°C on 25 February 2022.
2. The condensing water pump (CWP-09) has been ramped down its VSD speed from 48Hz to 45 Hz on 25 February 2022.
3. Due to the cold outdoor air temperature, electrical heaters in AHUs at north-east and north-west entrance of Casino have been operated between 1 – 11 February 2022 and 20 – 27 February 2022.
4. In order to provide domestic hot water for tower and space heating in podium and tower, large steam boiler has been operated between 1 – 11 February 2022 and 20 – 27 February 2022.

Cotai

1. The chilled water supply temperature setpoint has been finetuned from the temperature range of 7°C – 8°C to 7.5°C – 8.5°C on 25 February 2022.
2. All eight (8) cooling tower fans are shut off and one (1) condensing water circuit is maintained for circulation on 25 February 2022.
3. Two (2) condensing water pump (CWP-5 and CWP-6) VSD speed has been ramped down to 31 Hz on 25 February 2022, which can maintain the minimum condensing water flow rate for Genset and one (1) chiller.
4. Due to the temporary closure of L1 gaming area, the below practices have been proceeded.
    a. Indoor air temperature setpoints have been increased and the damper positions have been finetuned as minimum position for maintaining basic ventilation in Pit 201 and Pit 202 on 9 December 2021. The electric duct heaters at VAV boxes have been turned off automatically and the supply air fan motor speed (AH-3F-11 and AH-6M-5) have been reduced by 15 – 25%.
    b. The supply air temperature setpoints of AH-3F-11 for Pit 201 and AH-6M-5 for Pit 202 have been increased from 16°C to 18.5°C on 9 December 2021.
    c. Decorative chandeliers in rooms and open area of Pit 202 and Pit 203 have been switched off on 10 December 2021.
    d. Heaters at balconies of Pit 202 have been switched off on 10 December 2021.
5. Two (2) 95kW electric heaters in the calorifiers have been switched on for space heating from the week of 20 December 2021.
6. Due to the cold outdoor temperature between 20 – 25 February 2022, the natural gas boilers have been switched on for space heating. The boilers are then switched off in the morning of 25 February 2022.
7. Two (2) split type air conditioners at Genset room have been switched on starting from 20 February 2022 due to cold weather.

Consumption Intensity

The YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m^2)	% Change Compared to 2021
Macau	21.4	**+6.5%**
Cotai	21.2	**+0.4%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m^3/ m^2)	% Change Compared to 2021
Macau	0.19	**-3.0%**
Cotai	0.12	**-4.4%**

Note: Cotai Area: 374,138 m^2 Macau Area: 211,545 m^2

ISO 50001 Certification Audit

Macau

The 2022 internal audit has been completed on 4 March 2022 and discovered three (3) "Areas for Improvement". The next surveillance audit will be scheduled on 30 March 2022.

Cotai

The 2021 internal audit has been completed on 26 November 2021 and discovered three (3) "Areas for Improvement".

Target Milestones for the Year 2022

The following activities are planned for completion this year:

- To perform ISO 50001 2022 Internal and External Audit for MGM Macau and MGM Cotai
- To deploy and fine-tune the DGLux dashboard with heatmap development extension
- To perform 2021 Carbon and Waste Audit for MGM Macau and MGM Cotai
- To assist MGM with the COTAI phase 2 design and planning
- To investigate and implement the below list of projects during the Year 2022

Table 5. MGM Macau Year 2022 Projects

Item	Initiatives
Electricity	
1	Chilled Water Pump Retrofit (Phase 1)
2	AHU Static Pressure Set-Point Reset (Phase 2)
3	PICCV Implementation (Phase 3)
4	Air Side Equipment EC Fan/Motor Replacement
5	VAV Space Temperature Setpoint Optimization with Heatmap Control
6	Heatmap Phasing Implementation (Stage 3, 9a, 9b, 10a)
7	Neon Light LED Replacement (Phase 2)
Water	
8	CT Conductivity Blowdown Optimization

Table 6. MGM Cotai Year 2022 Projects

Item	Initiatives
Electricity	
1	Chiller Plant Control Optimization (CHWP/CDWP/CT)
2	Hot Water System Upgrade 1. Increase Max. No of Plate for Pre-heat Compact Unit 2. Upgrade Booster Compact Unit 3. Install Electric Heater for Booster Calorifier
3	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)
4	PICCV Implementation (Phase 3)
5	Hotel Towers Building Pressurization and EAF Optimization
6	Kitchen Exhaust Fan (KEF) Steps Speed Control Optimization
7	Carpark Ventilation Fans Control Optimization with IoT Sensor Installation (Phase 2)
8	VAV Space Temperature Setpoint Optimization with Heatmap Control
9	Heatmap Phasing Implementation (Stage 5, 6, 7, 8a, 8b, 10b)
10	Centralized RO Pump Set Replacement (10 Units)
Water	
11	Extension of CDP Water Recycling
12	Rainwater Harvesting for CT Make Up (Phase 1)
13	CT Conductivity Blowdown Optimization

TABLE OF CONTENT

EXECUTIVE SUMMARY 2
ISO 50001 Certification Audit 4
Target Milestones for the Year 2022 4
1. Portfolio Performance Summary 8
1.1 Electricity Use Intensity (EUI) 8
1.2 Water Use Intensity (WUI) 9
2. MGM Macau Performance 10
2.1 Electricity Consumption 10
2.2 LPG Consumption 12
2.3 ISO50001 Energy Consumption Record 14
2.4 Water Consumption 15
2.5 Macau Year 2022 Projects 17
3. MGM Cotai Performance 18
3.1 Electricity 18
3.2 Natural Gas Consumption 21
3.3 ISO50001 Energy Consumption Record 23
3.4 Water Consumption 24
3.5 Cotai Year 2022 Projects 26
Appendix A 27
Regression Analysis 27
MGM Macau 27
MGM Cotai 27

ACKNOWLEDGEMENT

Energenz has been working with MGM Team as the energy manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that Energenz has assisted MGM Macau and Cotai to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, and China 3 Star Green Building Operation Label.

Energenz appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

1. Portfolio Performance Summary

1.1 Electricity Use Intensity (EUI)

Macau's electricity use intensity (EUI) in February 2022 had increased by **1.8%** compared to the same month in 2021 and increased by **6.5%** in 2022 compared to 2021. Meanwhile, Cotai's EUI in February 2022 has decreased by **1.4%** compared to the same month in 2021 and increased by **0.4%** in 2022 compared to 2021.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2021 Actual EUI (kWh/m²)	Year 2022 Actual EUI (kWh/m²)	% Change	Year 2021 Actual EUI (kWh/m²)	Year 2022 Actual EUI (kWh/m²)	% Change
January	20.4	22.6	+11.0%	21.8	22.3	+2.1%
February	19.8	20.2	+1.8%	20.4	20.1	-1.4%
March						
April						
May						
June						
July						
August						
September						
October						
November						
December						
Average	**20.1**	**21.4**	**+6.5%**	**21.1**	**21.2**	**+0.4%**

1.2 Water Use Intensity (WUI)

Macau's water use intensity (WUI) in February 2022 increased by **1.1%** compared to the same month in 2021 and decreased by **3.0%** in 2022 compared to 2021. Meanwhile, Cotai's WUI in February 2022 decreased by **10.9%** compared to the same month in 2021 and decreased by **4.4%** in 2022 compared to 2021.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2021 Actual WUI (m^3/m^2)	Year 2022 Actual WUI (m^3/m^2)	% Change	Year 2021 Actual WUI (m^3/m^2)	Year 2022 Actual WUI (m^3/m^2)	% Change
January	0.19	0.18	-7.2%	0.116	0.119	+3.2%
February	0.19	0.20	+1.1%	0.14	0.12	-10.9%
March						
April						
May						
June						
July						
August						
September						
October						
November						
December						
Average	**0.19**	**0.19**	**-3.0%**	**0.13**	**0.12**	**-4.4%**

2. MGM Macau Performance

2.1 Electricity Consumption

The actual electricity consumption and cost in February 2022 increased by **1.8%** and **7.7%** respectively compared to the same month in 2021. To date, MGM Macau has increased the actual electricity consumption and associated costs by **6.5%** and **12.4%** respectively.

Table 9. Macau - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	4,310,160	4,784,280	+11.0%	4,717,260	5,519,350	+17.0%	+15.8%
February	4,190,760	4,266,240	+1.8%	4,591,170	4,942,590	+7.7%	-90.8%
March							
April							
May							
June							
July							
August							
September							
October							
November							
December							
Total/Average	**8,500,920**	**9,050,520**	**+6.5%**	**9,308,430**	**10,461,940**	**+12.4%**	**-60.6%**

Important note: *Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization.*

Energenz summarized the key items which could result in electricity consumption increase below:

- Air side equipment electricity consumption has increased 20.1% due to the AHU electrical heater was operated for space heating in majority of the February 2022.
- Lift & escalator electricity consumption has increased 14.1% due to the occupancy increased by 22.4%

Table 10 presents the comparison of electricity rates in the years 2021 and 2022. The electricity rate in February 2022 increased by **5.7%** compared to the same month in 2021.

Table 10. Macau - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2021 – Electricity Rate (MOP/kWh)	Year 2022 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.09	1.15	+5.4%
February	1.10	1.16	+5.7%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**1.09**	**1.16**	**+5.6%**

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in February 2022 decreased by **7.9%** compared to 2019 normalized consumption in the same month.

Table 11. Macau - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2022 Actual Electricity Consumption (kWh)	% Change - Normalized
January	4,790,450	4,784,280	-0.1%
February	4,631,715	4,266,240	-7.9%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**9,422,165**	**9,050,520**	**-3.9%**

2.2 LPG Consumption

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, Energenz focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption and cost in February 2022 increased by **96.0%** and **129.2%** respectively compared to the same month in 2021. To date, MGM Macau has increased the actual LPG consumption by **36.5%** and associated costs increased by **75.7%**.

Table 12. Macau - Summary of Actual Monthly LPG Consumption and Cost

Monthly Period	Actual LPG Consumption (kg)			Actual LPG Cost (MOP)			% Change – Min OAT
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	121,398	121,221	-0.1%	947,708	1,327,490	+40.1%	+56.0%
February	74,797	146,623	+96.0%	632,778	1,450,103	+129.2%	-52.6%
March							
April							
May							
June							
July							
August							
September							
October							
November							
December							
Total/Average	**196,195**	**267,844**	**+36.5%**	**1,580,486**	**2,777,593**	**+75.7%**	**-16.7%**

Important note: *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

Energenz summarized the key items which could result in LPG consumption increase below:

- Boilers LPG consumption increased by 165.3% in February 2022 and min OAT has decreased by 52.6% in February 2022 compared with the same month in 2021.
- Large boiler has been operated for 19 days in February 2022, however it was not operated in February 2021.
- Hotel upper zone domestic hot water and restaurant LPG consumption increased by 30.5% and 5.5% in February 2022 respectively compared with the same month in 2021.

Table 13 presents the comparison of LPG rates in the years 2021 and 2022. The LPG rate in February 2022 increased by **16.9%** compared with the same month in 2021.

Table 13. Macau - Summary of Actual Monthly LPG Rate

Monthly Period	Year 2021 – LPG Rate (MOP/kg)	Year 2022 – LPG Rate (MOP/ kg)	% Change – LPG Rate
January	7.81	10.95	+40.3%
February	8.46	9.89	+16.9%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**8.13**	**10.42**	**+28.1%**

Table 14 summarizes the monthly normalized LPG consumption. The actual LPG consumption in February 2022 increased by **24.5%** compared to 2019 normalized consumption in the same month.

Table 14. Macau - Summary of Normalized Monthly LPG Consumption

Monthly Period	2019 Normalized LPG Consumption (kg)	2022 Actual LPG Consumption (kg)	% Change – Normalized
January	95,134	121,221	+27.4%
February	117,728	146,623	+24.5%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**212,863**	**267,844**	**+25.8%**

2.3 ISO50001 Energy Consumption Record

The Macau ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to February 2022 is **4.8%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to the 2019 baseline.

Table 15. Macau ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to February 2022	2023	2024	2025
Macau	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-5.9%**	**-4.8%**	-	-	**-**

Table 16. Macau ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2022 Actual Energy Consumption (GJ)	% Change – Normalized vs Actual
January	21,555	22,715	+5.4%
February	22,007	22,000	-0.03%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**43,562**	**44,715**	**+2.6%**

Important note: *Factors used for fuels' unit conversion: 1) Electricity (0.0036) and LPG (0.0453)*

2.4 Water Consumption

Based on the existing water metering system, the overall site water consumption can be broken down into four (4) categories: Kitchen and Restaurants; Hotel/Tower; Fire Services; and Chiller Plant & miscellaneous[2].

According to Table 17, the actual water consumption in February 2022 increased by **1.1%** compared to the same month in 2021. To date, MGM Macau has decreased actual water consumption and associated costs by **3.0%** and **1.2%** respectively.

Table 17. Macau - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2021	2022	% Change Actual	2021	2022	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	39,952	37,078	-7.2%	299,330	287,500	-4.0%	+15.8%	+36.7%
February	41,079	41,524	+1.1%	317,780	322,270	+1.4%	-90.8%	+43.2%
March								
April								
May								
June								
July								
August								
September								
October								
November								
December								
Total/Average	**81,031**	**78,602**	**-3.0%**	**617,110**	**609,770**	**-1.2%**	**-60.6%**	**+40.2%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption increase below:

- Kitchen water consumption for February 2022 increased by 16.2% compared to the same month in 2021. Moreover, the food cover increased by 43.2% compared to the same month in 2021.

[2] Miscellaneous includes the water consumption from toilets, laundry, etc.

Table 18 presents the comparison of water rates in the years 2021 and 2022. The water rate in February 2022 increased by **0.3%** compared to the same month in 2021. The average utility rate of water has increased by **1.9%** compared to 2021.

Table 18. Macau - Summary of Actual Monthly Water Rates

Monthly Period	Year 2021 – Water Rate (MOP/m^3)	Year 2022 – Water Rate (MOP/m^3)	% Change – Water Rate
January	7.49	7.75	+3.5%
February	7.74	7.76	+0.3%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**7.61**	**7.76**	**+1.9%**

Table 19 summarized the monthly normalized water consumption. The actual water consumption in February 2022 increased by **10.7%** compared to 2019 normalized consumption in the same month.

Table 19. Macau - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m^3)	2022 Actual Water Consumption (m^3)	% Change - Normalized
January	35,691	37,078	+3.9%
February	37,518	41,524	+10.7%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**73,209**	**78,602**	**+7.4%**

2.5 Macau Year 2022 Projects

Energenz will continue to perform analysis and coordinate with MGM to implement the following projects in 2022 and reduce the site energy and water consumption which supports Energenz annual normalized energy savings target.

Table 20. MGM Macau Year 2021 Projects

Item	Initiatives
Electricity	
1	Chilled Water Pump Retrofit (Phase 1)
2	AHU Static Pressure Set-Point Reset (Phase 2)
3	PICCV Implementation (Phase 3)
4	Air Side Equipment EC Fan/Motor Replacement
5	VAV Space Temperature Setpoint Optimization with Heatmap Control
6	Heatmap Phasing Implementation (Stage 3, 9a, 9b, 10a)
7	Neon Light LED Replacement (Phase 2)
Water	
8	CT Conductivity Blowdown Optimization

3. MGM Cotai Performance

3.1 Electricity

The actual electricity consumption in February 2022 decreased by **1.4%** and cost increased by **3.8%** compared to the same month in 2021. To date, MGM Cotai has increased actual electricity consumption and associated costs by **0.4%** and **5.7%** respectively.

Table 21. Cotai - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	8,168,400	8,336,400	+2.1%	9,315,910	10,013,460	+7.5%	+15.8%
February	7,637,640	7,528,560	-1.4%	8,737,740	9,066,060	+3.8%	-90.8%
March							
April							
May							
June							
July							
August							
September							
October							
November							
December							
Total/Average	**15,806,040**	**15,864,960**	**+0.4%**	**18,053,650**	**19,079,520**	**+5.7%**	**-60.6%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization. The e-Bus electricity consumption has been included with the presented electricity consumption above.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- February 2022 Chiller plant electricity consumption has decreased 53.0% due to a 90.8% CDD decrease when compared with the same month last year.

Table 22 presents the monthly e-bus electricity consumption in MGM Cotai. The e-bus electricity consumption in February 2022 increased by **47.2%** compared to the same month in 2021.

Table 22. Cotai - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2021	2022	% Change - Actual
January	44,839	42,287	-5.7%
February	32,534	47,905	+47.2%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**77,373**	**90,192**	**+16.6%**

Table 23 presents the comparison of electricity rates in the years 2021 and 2022. The electricity rate in February 2022 increased by **5.3%** compared to the same month in 2021.

Table 23. Cotai - Summary of Actual Monthly Electricity Rates

Monthly Period	Year 2021 – Electricity Rate (MOP/kWh)	Year 2022 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.14	1.20	+5.3%
February	1.14	1.20	+5.3%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**1.14**	**1.20**	**+5.3%**

Table 24 summarized the monthly normalized electricity consumption. The actual electricity consumption in February 2022 decreased by **10.3%** compared to 2019 normalized consumption in the same month.

Table 24. Cotai - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2022 Actual Electricity Consumption (kWh)	% Change - Normalized
January	8,635,397	8,336,400	-3.5%
February	8,394,419	7,528,560	-10.3%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**17,029,815**	**15,864,960**	**--6.8%**

3.2 Natural Gas Consumption

The actual natural gas consumption and cost in February 2022 increased by **40.3%** and **47.9%** respectively compared to the same month in 2021. To date, MGM Cotai has increased actual natural gas consumption by **17.9%** and the associated costs increased by **24.4%**.

Table 25. Cotai - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual Natural Gas Consumption (m^3)			Actual Natural Gas Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost		
January	60,643	59,378	-2.1%	363,990	376,240	+3.4%	+56.0%	+22.3%
February	54,221	76,089	+40.3%	325,930	482,130	+47.9%	-52.6%	+31.6%
March								
April								
May								
June								
July								
August								
September								
October								
November								
December								
Total/Average	**114,864**	**135,467**	**+17.9%**	**689,920**	**858,370**	**+24.4%**	**-16.7%**	**+27.1%**

Important note: *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization.*

Energenz summarized the key items which could result in natural gas consumption increase below:

- Due to the outdoor temperature drops between 20 – 25 February 2022, the natural gas boilers have been switched on for space heating. The boilers were switched off in the morning of 25 February 2022.

Table 26 presents the comparison of natural gas rates in the years 2021 and 2022. The natural gas rate in February 2022 increased by **5.4%** compared with the same month in 2021.

Table 26. Cotai - Summary of Actual Monthly Natural Gas Rates

Monthly Period	Year 2021 – Natural Gas Rate (MOP/m³)	Year 2022 – Natural Gas Rate (MOP/m³)	% Change – Natural Gas Rate
January	6.00	6.34	+5.6%
February	6.01	6.34	+5.4%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**6.01**	**6.34**	**+5.5%**

Table 27 summarized the monthly normalized natural gas consumption. The actual natural gas consumption in February 2022 decreased by **42.7%** compared to 2019 normalized consumption in the same month.

Table 27. Cotai - Summary of Normalized Monthly Natural Gas Consumption

Monthly Period	2019 Normalized Natural Gas Consumption (m³)	2022 Actual Natural Gas Consumption (m³)	% Change - Normalized
January	125,664	59,378	-52.7%
February	132,723	76,089	-42.7%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**258,387**	**135,467**	**-47.6%**

3.3 ISO50001 Energy Consumption Record

Cotai ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. Based on the official agreement with MGM Cotai, e-bus consumption should be included in the energy consumption tracking as part of the ISO50001 requirement. To date, normalized energy saving achieved up to February 2022 is **7.5%** as shown in Table 28Table 28. Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

Table 28. Cotai ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to February 2022	2023	2024	2025
Cotai	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-6.8%**	**-7.5%**	-	-	-

Table 29. Cotai ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2022 Actual Energy Consumption (GJ)	% Change - Normalized vs Actual
January	35,448	32,071	-9.5%
February	34,825	29,743	-14.6%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**70,273**	**61,815**	**-12.0%**

***Important note**: Factors used for fuels' unit conversion: 1) Electricity (0.0036) and Natural Gas (0.0347), e-bus consumption is included*

3.4 Water Consumption

According to Table 30, the actual water consumption in February 2022 decreased by **10.9%** compared to the same month in 2021. To date, MGM Cotai has decreased actual water consumption and associated costs by **4.4%** and **4.4%** respectively.

Table 30. Cotai - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2021	2022	% Change Actual	2021	2022	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	43,292	44,658	+3.2%	336,700	347,240	+3.1%	+15.8%	+22.3%
February	50,837	45,298	-10.9%	395,180	352,330	-10.8%	-90.8%	+31.6%
March								
April								
May								
June								
July								
August								
September								
October								
November								
December								
Total/ Average	**94,129**	**89,956**	**-4.4%**	**731,880**	**699,570**	**-4.4%**	**-60.6%**	**+27.1%**

***Important note**: Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Cooling tower makeup water consumption decreased by 52.6% due to 90.8% CDD decrease when compared with the same month last year.

Table 31 presents the comparison of water rates in the years 2021 and 2022. The water rate in February 2022 increased by **0.1%** compared to the same month in 2021. The average utility rate of water has increased by **0.02%** compared to 2021.

Table 31. Cotai - Summary of Actual Monthly Water Rates

Monthly Period	Year 2021 – Water Rate (MOP/m³)	Year 2022 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.78	7.78	-0.02%
February	7.77	7.78	+0.1%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**7.78**	**7.78**	**+0.02%**

Table 32 summarized the monthly normalized water consumption. The actual water consumption in February 2022 decreased by **7.6%** compared to 2019 normalized consumption in the same month.

Table 32. Cotai - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2022 Actual Water Consumption (m³)	% Change - Normalized
January	48,361	44,658	-7.7%
February	49,015	45,298	-7.6%
March			
April			
May			
June			
July			
August			
September			
October			
November			
December			
Total/Average	**97,376**	**89,956**	**-7.6%**

3.5 Cotai Year 2022 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2022 and reduce the site energy and water consumption which supports Energenz annual normalized energy savings target.

Table 33. MGM Cotai Year 2022 Projects

Item	Initiatives
Electricity	
1	Chiller Plant Control Optimization (CHWP/CDWP/CT)
2	Hot Water System Upgrade 1. Increase Max. No of Plate for Pre-heat Compact Unit 2. Upgrade Booster Compact Unit 3. Install Electric Heater for Booster Calorifier
3	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)
4	PICCV Implementation (Phase 3)
5	Hotel Towers Building Pressurization and EAF Optimization
6	Kitchen Exhaust Fan (KEF) Steps Speed Control Optimization
7	Carpark Ventilation Fans Control Optimization with IoT Sensor Installation (Phase 2)
8	VAV Space Temperature Setpoint Optimization with Heatmap Control
9	Heatmap Phasing Implementation (Stage 5, 6, 7, 8a, 8b, 10b)
10	Centralized RO Pump Set Replacement (10 Units)
Water	
11	Extension of CDP Water Recycling
12	Rainwater Harvesting for CT Make Up (Phase 1)
13	CT Conductivity Blowdown Optimization

Appendix A

Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target of 4% accurately for MGM Macau and MGM Cotai. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2022.

MGM Macau

Utilities	2022	
Electricity (kWh)	CDD (Base T=17°C)	
LPG (kg)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM Cotai

Utilities	2022	
Electricity (kWh)	CDD (Base T=17°C)	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

Generating Great Returns.
Improving The Environment.

